

VAISHNAVI PUJALA

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SUMMARY

Computer Science professional with a genuine passion for software engineering and full-stack development. Combines hands-on engineering experience across web, backend, and distributed systems with published research in applied AI, bringing both the technical depth to design scalable solutions and the analytical mindset to solve complex engineering problems

EDUCATION

Georgia State University, Atlanta, GA, USA | M.S. Computer Science, GPA: 3.3/4.0 Aug 2024 – May 2026
Coursework: Machine Learning, Neural Networks, Deep Learning, LLMs, Federated Learning
G. Narayanamma Institute of Technology and Science, India | B.Tech Computer Science, GPA: 3.4/4.0 Aug 2020 – Jul 2024

TECHNICAL SKILLS

Languages: Java, Python, JavaScript, TypeScript, Go, C, C++, SQL, HTML, CSS
Frameworks & Libraries: React.js, Next.js, Node.js, Angular, FastAPI, Flask, Django, Flutter, Tailwind CSS, PyTorch, TensorFlow, Keras
Cloud & DevOps: AWS (S3, SQS, SNS, Lambda), GCP, Docker, Kubernetes, GitHub Actions, CI/CD, Linux, Bash
Databases: PostgreSQL, MongoDB, DynamoDB, Redis, NoSQL
Tools & Concepts: REST APIs, Microservices, System Design, OOP, Unit Testing, Agile, Kafka, NLP, LLMs, Tableau, Power BI, Data Structures

EXPERIENCE

- AI Developer Intern** | NeuroHeart AI Labs LLC, USA Jun 2026 – Present
- Developing agentic AI and LLM-powered applications for live client engagements, architecting conversational flows and knowledge base pipelines across the full requirements-to-deployment cycle
 - Integrating generative AI APIs and retrieval-augmented workflows into production-ready modules, collaborating cross-functionally on debugging, testing, and deployment for client-facing features
- Graduate Research Assistant** | Georgia State University, USA Aug 2025 – May 2026
- Designed a privacy-preserving federated HAR system on the WISDM dataset using a CNN-BiLSTM backbone with Temporal Self-Attention and Domain Adversarial Neural Networks (DANN), achieving 70.48% cross-domain accuracy (Phone → Watch transfer), an 8.7% improvement over the no-adaptation baseline
 - Benchmarked 5 domain adaptation strategies (Source-Only, Centralized DANN, Federated DANN via FedProx, Attention-augmented DANN, and Personalized Federated DANN) using Gradient Reversal Layer scheduling and per-client personalized heads to maximize target-domain generalization while keeping all raw sensor data on local devices
- Graduate Teaching and Lab Assistant** | Georgia State University, USA Aug 2024 – May 2025
- Instructed and mentored 100+ undergraduate students in CSC3320 System-Level Programming and Python-based data analytics, delivering hands-on coding exercises and lab sessions on C, C++, memory management, pointers, and data visualization
 - Collaborated with faculty to develop lab materials and regression/classification exercises, analyzing student performance data to refine assessment strategies and improve learning outcomes
- Computer Vision & ML Engineer** | Atal Incubation Centre, AIC-GNITS Foundation, India Jul 2023 – Jul 2024
- Constructed and optimized a computer vision pipeline for copy-move image forgery detection using CNN and VGG-16, handling preprocessing, feature extraction, and model training in Python, TensorFlow/Keras, and OpenCV to reach 90% detection accuracy
 - Delivered a full-stack ML inference system by integrating a Flask-based backend with an interactive HTML/CSS/JavaScript frontend, enabling real-time image upload and forgery localization, featured in Springer ICCCE 2024
- AI/ML Research Associate** | Innovation and Incubation Cell - GNITS, India Aug 2022 – Jun 2023
- Developed a heart disease prediction system comparing Random Forest, Naïve Bayes, and Logistic Regression, applying feature selection, hyperparameter tuning, and cross-validation to achieve 91% accuracy, published in Springer ICDSMLA 2023
 - Performed exploratory data analysis and produced interactive visualizations using Plotly, Seaborn, and Matplotlib to surface health risk patterns and support integration of predictive models into AI-driven healthcare analytics workflows

PROJECTS

- CodeSentinel** | Python, FastAPI, LangGraph, LLaMA 3, React, PostgreSQL, Redis, Docker Mar 2026 – May 2026
- Engineered a multi-agent security scanner using LangGraph to orchestrate 4 parallel agents (SAST, Secrets, Dependency, Fix) analyzing GitHub repositories for vulnerabilities, with LLaMA 3 fix generation, Celery + Redis async pipeline, WebSocket progress streaming, and JWT-authenticated FastAPI endpoints persisting results to PostgreSQL
- Hospital Management System** | Java, Spring Boot, React, TypeScript, PostgreSQL, Kafka, Redis, Docker Dec 2025 – Feb 2026
- Built a full-stack enterprise Hospital Management System with Java 17 and Spring Boot 3, implementing JWT-based role-based access control across 3 user roles (Admin, Doctor, Patient) with dedicated React TypeScript dashboards, real-time appointment booking, and Swagger/OpenAPI documentation
- Distributed Task Queue System** | Python, FastAPI, Redis, PostgreSQL, React.js, WebSockets, Docker Sep 2025 – Nov 2025
- Architected a Celery-like task queue with Redis-backed priority queues, 3 async workers, exponential backoff retries, dead letter queue, and a real-time React dashboard over WebSockets, containerized with Docker, processing 1,000+ tasks per minute under load

PUBLICATIONS

- Springer Journal:** [An effective Prediction of Heart Diseases using Machine Learning Techniques](#), ICDSMLA 2023 – Machine Learning
- Springer Journal:** [An approach to Detect Copy Move Forgery Detection using Deep Learning Techniques](#), ICCCE 2024 – Computer Vision